Lab3: Web application with MongoDB.

v. 27/05/2025 – EPF – A. Gademer & E. Fragale

Abstract:

- Design the backend of a web application with MongoDB.
- Comparison with classical relational database.

⚠️ **Important remark**: This subject will use a set of icon and graphical chart to indicate several levels of information.

- 👉: Something that you should **do** to complete the Lab
- 💾: You need to **validate** something (call the teacher, save something for deposit)
- 💡: A tips or **useful** remark to help you in your lab
- 🧠: An explanation on a new knowledge, that you should probably know/**retain**
- ⚠️: Something important that you **need** to read
- ⭐: Something to try if you want to **explore** further on the subject
- IT SHOULD NOT BE NEEDED TO BE WRITTEN BUT EVERYTHING ON WHITE BACKGROUND SHOULD BE READ ALSO! (Only ⭐ is optional).

Preamble:

For this lab, you will need a **working Docker installation** (see previous Lab).

Important note:

This Lab will continue the work done in the *Tutorial 2: Case study. Database design from existing data*. **You will keep the same groups** (see list on Moodle).

Exercise 1: Back to the drawing board.

👉 Reopen the Excel file from the Tutorial 2. Reopen the Looping file and all the files that you submitted at the time.

[Reminder] User story 😊 :

During the specification meeting, the client has shown you the Excel file that he uses for his task. He wants you to develop a web application to replace it.
New functionalities:

- User management: each user (identified by its email address), should only see the data meant for they (neutral pronoun).
- One common database instead of one excel file for each semester.

[New constraints] User story:

- The sister-in-law of the client has told her that MongoDB was the new chic. She wants a MongoDB database.
- Your CTO want to take the chance to test MongoDB but wants to determine the real interest of the tool, she ask you an internal report showing the advantages / disadvantages of your MongoDB implementation.

Objectives of this lab:

- As part of the red team, your goal is to design and implement the backend of the client web application with MongoDB. (The frontend will be developed by the blue team.)
- The web app goal is to manage the skill appreciations of the EPF students during their 4th and 5th semester in the MDE program but in this Lab you will only implement a proof-of-concept for the 2 first Education Units (UE1 & UE2) and for the student list that was provided to you (with the constraint of the Exchange students).
- The list of skill, associated teacher, student list (even the grade scale) can change from one semester to the other. But the global principle can be considered stable (for the moment 😜): One or several teachers can assign an appreciation for each student of their class for each of the skills associated to their module. The appreciation is selected in an enumeration. Each appreciation is associated to a numerical mark. The numerical mark of the module is the pondered sum of the appreciation obtained (the ponderation factor is given for each skill).
- Note: for several reasons (minor groups, exchange students, dropouts, students resitting from previous year…), some specific student may be dispensed of some modules. *You do not need to keep the information of why they are dispensed*.

Deliverables:

- An oral presentation that explains your design choices:
  - Which collection(s) and why.
  - How does it answer to the client need.

💾 : The oral presentation will take place TOMORROW. It should be ready tonight.

- A BSON export of the collection(s) corresponding to the 2 Education Units
- A list of MongoDB query for the blue team to be integrated in the frontend:
  - List all the semesters associated with the Teacher T.
  - List all potential appreciation levels associated with the semester S.
  - List all skills associated with the Teacher T for the semester S.
  - List all skills associated with the Teacher T for the semester S that are not yet published.
  - Teacher T update the appreciation of the skill Sk of the semester S with the value V.
  - Teacher T update the status of the skill Sk with the value V.
  - List all the semesters associated with the Student St.
  - List (all the field but the appreciations) of all the skills associated with the Student St.
  - List the appreciations of all the published skills associated with the Student St.

⚠️ Each query should come with a test query (chosen arguments and the corresponding expected result). **Better less queries but well documented**.

- A report on the advantage/inconvenient of the MongoDB implementation over the MariaDB one.
- (Bonus) The description of the algorithms (in pseudo-code or in programming language) of the following functions:
  - UpdateAppreciation(Teacher T, Semester S, Skill sk, Value V)
  - ShowStudentAppreciations(Student St, Semester S)
  - ShowStudentNumericalGrade(Student St, Semester S, Module M)

💾 : The following deliverables will be deposit on Moodle for **Sunday midnight**.